# RELATIONAL ALGEBRA EXERCISES
# SEM I – 2025/2026

## QUESTION 1

Relational Schema:

BOOK (Bid, Title, Year, Category, Lid)
MEMBER (Mid, MName, City)
BORROW (Mid, Bid, DateBorrowed)
LIBRARIAN (Lid, LName, Branch)

**BOOK**

| Bid | Title | Year | Category | Lid |
|---|---|---|---|---|
| 1 | AI Revolution | 2021 | Technology | 1 |
| 2 | History of Art | 2019 | Art | 2 |
| 3 | World Atlas | 2020 | Geography | 1 |

**MEMBER**

| Mid | MName | City |
|---|---|---|
| 1 | James | Johor Bahru |
| 2 | Aisha | Kuala Lumpur |
| 3 | Lim | Penang |

**LIBRARIAN**

| Lid | LName | Branch |
|---|---|---|
| 1 | Ravi | Main Branch |
| 2 | Sarah | Art Wing |
| 3 | Ismail | Southern Key |

**BORROW**

| Mid | Bid | DateBorrowed |
|---|---|---|
| 1 | 1 | 2024-02-10 |
| 2 | 2 | 2024-03-15 |
| 3 | 3 | 2024-04-01 |

a) List all book titles and their categories.

b) List titles of books borrowed by members from 'Johor Bahru'.

c) Find librarians who do not manage any books.

d) List names of members who borrowed books published after 2020.

e) List librarian names and the titles of books they manage.

f) Product output for the given relational algebra:

i) $\pi_{Title,Year}$ (BOOK)

ii) $\pi_{Title,Category}$ ($\sigma_{Branch='MainBranch'}$(LIBRARIAN $\bowtie_{Lid=Lid}$ BOOK))

iii) $\pi_{MName,Title}$ (MEMBER$\bowtie_{Mid=Mid}$ (BORROW$\bowtie_{Bid=Bid}$ BOOK))

iv) πLName,Title,Branch (LIBRARIAN⋈Lid=Lid BOOK)

a) List all book titles and their categories.

$\pi_{Title, Category}(Book)$

| Title | Category |
|---|---|
| AI Revolution | Technology |
| History of Art | Art |
| World Atlas | Geography |

b) List titles of books borrowed by members from 'Johor Bahru'.

$\pi_{titles}(\sigma_{city="Johor Bahru"}(Member) \bowtie Borrow \bowtie Book)$

**Member**

| Mid | city |
|---|---|
| 1 | Johor Bahru |

$\bowtie$

**Borrow**

| Mid | Bid |
|---|---|
| 1 | 1 |

$\bowtie$

**Book**

| Bid | Title |
|---|---|
| 1 | AI Revolution |

Title = AI Revolution

c) Find librarians who do not manage any books.

$\pi_{LName}(\pi_{Lid}(LIBRARIAN) - \pi_{Lid}(BOOK) \bowtie LIBRARIAN)$

| Lid |
|---|
| 1 |
| 2 |
| 3 |

$-$

| Lid |
|---|
| 1 |
| 2 |

$=$

| Lid |
|---|
| 3 |

$\longrightarrow$

| Lid | LName | Branch |
|---|---|---|
| 3 | Ismail | Southern Key |

LName = Ismail

d) List names of members who borrowed books published after 2020.

$\pi_{MName}(\sigma_{Year>2020}(Book) \bowtie (Borrow) \bowtie Member)$

| Bid | Year |
|---|---|
| 1 | 2021 (Pass) |
| ~~2~~ | ~~2019~~ (Fail) |
| ~~3~~ | ~~2020~~ (Fail) |

$\bowtie$

| Bid | Mid |
|---|---|
| 1 | 1 |

$\bowtie$

| Mid | MName |
|---|---|
| 1 | James |

List names = James

e) List librarian names and the titles of books they manage.

$\pi_{LName, Title}(LIBRARIAN \bowtie Book)$

| Lid | LName |
|---|---|
| 1 | Ravi |
| 2 | Sarah |
| 3 | Ismail |

$\bowtie$

| Lid | Title |
|---|---|
| 1 | AI Revolution |
| 2 | History of Art |
| 1 | World Atlas |

=

| LName | Lid | Title |
|---|---|---|
| Ravi | 1 | AI Revolution |
| Ravi | 1 | World Atlas |
| Sarah | 2 | History of art |

f) Product output for the given relational algebra:

i) $\pi_{Title,Year}$ (BOOK)

| Title | Year |
|---|---|
| AI Revolution | 2021 |
| History of Art | 2019 |
| World Atlas | 2020 |

ii) $\pi_{Title,Category}(\sigma_{Branch='MainBranch'}(LIBRARIAN \bowtie_{Lid=Lid} BOOK))$

| Lid | Branch | |
|---|---|---|
| 1 | Main Branch | (Pass) |
| ~~2~~ | ~~Art Wing~~ | (Fail) |
| ~~3~~ | ~~Souther Key~~ | (Fail) |

⋈

| Lid | Title | Category | |
|---|---|---|---|
| 1 | AI Revolution | Technology | (Pass) |
| ~~2~~ | ~~History of art~~ | ~~Art~~ | (Fail) |
| 1 | World Atlas | Geography | (Pass) |

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| Branch | LIBRARIAN.LID | BOOK.LID | Title | Category |
|---|---|---|---|---|
| Main Branch | 1 | 1 | AI Revolution | Technology |
| Main Branch | 1 | 1 | World Atlas | Geography |

iii) $\pi_{MName,Title}(MEMBER \bowtie_{Mid=Mid}(BORROW \bowtie_{Bid=Bid} BOOK))$

**Borrow**

| Mid | Bid |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |

⋈

**Book**

| Bid | Title |
|---|---|
| 1 | AI Revolution |
| 2 | History of Art |
| 3 | World Atlas |

=

| Mid | Borrow.Bid | Book.Bid | Title |
|---|---|---|---|
| 1 | 1 | 1 | AI Revolution |
| 2 | 2 | 2 | History of Art |
| 3 | 3 | 3 | World Atlas |

⋈

**Member**

| Mid | Mname |
|---|---|
| 1 | James |
| 2 | Aisha |
| 3 | Lim |

=

| Borrow.Mid | Title | Member.Mid | Mname |
|---|---|---|---|
| 1 | AI Revolution | 1 | James |
| 2 | History of Art | 2 | Aisha |
| 3 | World Atlas | 3 | Lim |

iv) πLName,Title,Branch (LIBRARIAN⋈Lid=Lid BOOK)

**Librarian**

| Lid | LName | Branch |
|---|---|---|
| 1 | Ravi | Main Branch (Pass) |
| 2 | Sarah | Art Wing (Pass) |
| ~~3~~ | ~~Ismail~~ | ~~Souther Key~~ (Fail) |

⋈Lid=Lid

**Book**

| Lid | Title |
|---|---|
| 1 | AI Revolution |
| 2 | History of Art |
| 1 | World Atlas |

=

| Librarian.Lid | LName | Branch | Book.Lid | Title |
|---|---|---|---|---|
| 1 | Ravi | Main Branch | 1 | AI Revolution |
| 2 | Sarah | Art Wing | 2 | History of Art |
| 1 | Ravi | Main Branch | 1 | World Atlas |

## QUESTION 2

Relational Schema:

PRODUCT (Pid, PName, Category, Price, Sid)
SUPPLIER (Sid, SName, City)
CUSTOMER (Cid, CName, City)
PURCHASE (Cid, Pid, DatePurchase)

**PRODUCT**

| Pid | PName | Category | Price | Sid |
|---|---|---|---|---|
| 1 | Rice | Grocery | 30 | 2 |
| 2 | Soap | Toiletry | 10 | 1 |
| 3 | Milk | Dairy | 8 | 1 |

**SUPPLIER**

| Sid | SName | City |
|---|---|---|
| 1 | Fresh Supply | Kuala Lumpur |
| 2 | Agro Mart | Johor Bahru |

**CUSTOMER**

| Cid | CName | City |
|---|---|---|
| 1 | Amir | Penang |
| 2 | Lina | Johor Bahru |
| 3 | Tan | Kuala Lumpur |

a) List all product names and their prices.

b) List names of suppliers located in 'Kuala Lumpur'.

c) Find suppliers who do not supply any products.

d) List product names bought by customers from 'Penang'.

e) List product names and supplier names together.

a) List all product names and their prices.

$\pi_{PName, Price}(PRODUCT)$

| PName | Price |
|---|---|
| Rice | 30 |
| Soap | 10 |
| Milk | 8 |

b) List names of suppliers located in 'Kuala Lumpur'.

$\pi_{SNames}(\sigma_{city = Kuala\ Lumpur}(SUPPLIER))$

| SName | City |
|---|---|
| Fresh Supply | Kuala Lumpur (Pass) |
| ~~Agro Mart~~ | ~~Johor Bahru~~ (Fail) |

c) Find suppliers who do not supply any products.

$\pi_{sname}(\pi_{sid}(SUPPLIER) - \pi_{sid}(PRODUCT) \bowtie SUPPLIER)$

**SUPPLIER**

| sid |
|---|
| 1 |
| 2 |

$-$

**PRODUCT**

| sid |
|---|
| 1 |
| 2 |

$=$

| sid |
|---|
| NULL |

d) List product names bought by customers from 'Penang'.

$\pi_{Pname}(\sigma_{city=Penang}(CUSTOMER))$

Error Question "PURCHASE" table is not provided

e) List product names and supplier names together.

$\pi_{PName, sName}(Product \bowtie_{sid=sid} SUPPLIER)$

**Product**

| Sid | PName |
|---|---|
| 2 | Rice |
| 1 | Soap |
| 1 | Milk |

$\bowtie_{sid=sid}$

**SUPPLIER**

| Sid | SName |
|---|---|
| 1 | Fresh Supply |
| 2 | Agro Mart |

=

| Product.sid | PName | SUPPLIER.Sid | SName |
|---|---|---|---|
| 2 | Rice | 2 | Agro Mart |
| 1 | Soap | 1 | Fresh Supply |
| 1 | Milk | 1 | Fresh Supply |

## QUESTION 3

Relational Schema:

FLIGHT (Fid, FName, Origin, Destination, Aid)
AIRPORT (Aid, AName, City)
PASSENGER (Pid, PName, Nationality)
BOOKING (Pid, Fid, SeatNo)

**FLIGHT**

| Fid | FName | Origin | Destination | Aid |
|---|---|---|---|---|
| 1 | AK101 | Kuala Lumpur | Tokyo | 1 |
| 2 | MH204 | Penang | Seoul | 2 |
| 3 | SQ555 | Singapore | London | 3 |

**AIRPORT**

| Aid | AName | City |
|---|---|---|
| 1 | KLIA | Kuala Lumpur |
| 2 | Penang Airport | Penang |
| 3 | Changi | Singapore |

**PASSENGER**

| Pid | PName | Nationality |
|---|---|---|
| 1 | Hiro | Japan |
| 2 | Siti | Malaysia |
| 3 | John | UK |

a) List all flight names and their destinations.

b) List flight names that depart from 'Kuala Lumpur'.

c) Find airports that have no flights departing.

d) List names of passengers who booked flights to 'Tokyo'.

e) List flight names and airport names they belong to.

Product output for the given relational algebra:

i) $\pi_{FName,Destination}$ (FLIGHT)

ii) $\pi_{FName,City}$ (FLIGHT$\bowtie_{Aid=Aid}$ AIRPORT)

iii) $\pi_{FName,Origin,Destination,Aname}$ (FLIGHT$\bowtie_{Aid=Aid}$ AIRPORT)

iv) $\pi_{AName,City}$ ($\sigma_{City='Penang'}$ (AIRPORT))

a) List all flight names and their destinations.

$\pi_{FName, Destination}(FLIGHT)$

| FName | Destination |
|---|---|
| AK101 | Tokyo |
| MH204 | Seoul |
| SQ555 | London |

b) List flight names that depart from 'Kuala Lumpur'.

$\pi_{FName}(\sigma_{Origin=Kuala\ Lumpur}(FLIGHT))$

| FName | Origin | |
|---|---|---|
| AK101 | Kuala Lumpur | (Pass) |
| ~~MH204~~ | ~~Penang~~ | (Fail) |
| ~~SQ555~~ | ~~Singapore~~ | (Fail) |

c) Find airports that have no flights departing.

$\pi_{AName}(\pi_{Aid}(AIRPORT) - \pi_{AID}(FLIGHT) \bowtie AIRPORT)$

**AIRPORT**

| AID |
|---|
| 1 |
| 2 |
| 3 |

$-$

**FLIGHT**

| AID |
|---|
| 1 |
| 2 |
| 3 |

$=$

| AID |
|---|
| NULL |

d) List names of passengers who booked flights to 'Tokyo'.

$\pi_{PName}(\sigma_{Destination=Tokyo}(FLIGHT) \bowtie Booking \bowtie PASSENGER)$

ERROR question. "BOOKING" table is not provided

e) List flight names and airport names they belong to.

$\pi_{FName, AName}(Flight \bowtie_{Aid=Aid} Airport)$

**FLIGHT**

| Aid | FName |
|---|---|
| 1 | AK101 |
| 2 | MH204 |
| 3 | SQ555 |

$\bowtie_{Aid=Aid}$

**AIRPORT**

| Aid | AName |
|---|---|
| 1 | KLIA |
| 2 | Penang Airport |
| 3 | Changi |

=

| FLIGHT.Aid | FName | AIRPORT.Aid | AName |
|---|---|---|---|
| 1 | AK101 | 1 | KLIA |
| 2 | MH204 | 2 | Penang Airport |
| 3 | SQ555 | 3 | Changi |

Product output for the given relational algebra:

i) $\pi_{FName, Destination}$ (FLIGHT)

$\pi_{FName, Destination}(FLIGHT)$

| FName | Destination |
|---|---|
| AK101 | Tokyo |
| MH204 | Seoul |
| SQ555 | London |

ii) $\pi_{FName,City}$ (FLIGHT$\bowtie_{Aid=Aid}$ AIRPORT)

**FLIGHT**

| FName | Aid |
|---|---|
| AK101 | 1 |
| MH204 | 2 |
| SQ555 | 3 |

$\bowtie_{Aid=Aid}$

**AIRPORT**

| AID | city |
|---|---|
| 1 | Kuala Lumpur |
| 2 | Penang |
| 3 | Singapore |

=

| FLIGHT.Aid | FName | AIRPORT.AID | city |
|---|---|---|---|
| 1 | AK101 | 1 | Kuala Lumpur |
| 2 | MH204 | 2 | Penang |
| 3 | SQ555 | 3 | Singapore |

iii) $\pi_{FName,Origin,Destination,Aname}$ (FLIGHT$\bowtie_{Aid=Aid}$ AIRPORT)

**FLIGHT**

| Aid | FName | Origin | Destination |
|---|---|---|---|
| 1 | AK101 | Kuala Lumpur | Tokyo |
| 2 | MH204 | Penang | Seoul |
| 3 | SQ555 | Singapore | London |

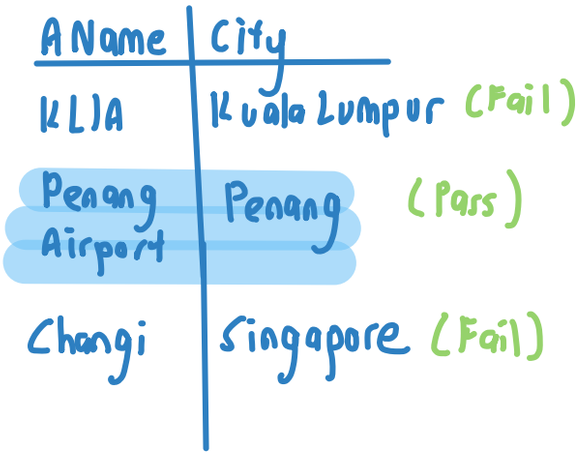
$\bowtie_{Aid=Aid}$

**AIRPORT**

| Aid | Aname |
|---|---|
| 1 | KLIA |
| 2 | Penang Airport |
| 3 | Changi |

| FLIGHT.Aid | FName | Origin | Destination | AIRPORT.Aid | Aname |
|---|---|---|---|---|---|
| 1 | AK101 | Kuala Lumpur | Tokyo | 1 | KLIA |
| 2 | MH204 | Penang | Seoul | 2 | Penang Airport |
| 3 | SQ555 | Singapore | London | 3 | Changi |

iv) $\pi_{AName,City}(\sigma_{City='Penang'}(AIRPORT))$



## QUESTION 4

Relational Schema:

COURSE (Cid, CName, Category, Year, Tid)
TEACHER (Tid, TName, Expertise)
STUDENT (Sid, SName, Country)
ENROLL (Sid, Cid, Grade)

**COURSE**

| Cid | CName | Category | Year | Tid |
|---|---|---|---|---|
| 1 | Python Basics | Programming | 2023 | 1 |
| 2 | Data Mining | Data Science | 2025 | 2 |
| 3 | Web Design | IT | 2024 | 1 |

**TEACHER**

| Tid | TName | Expertise |
|---|---|---|
| 1 | Dr. Ali | Programming |
| 2 | Ms. Tan | Data Science |
| 3 | Prof. John | Cybersecurity |

**STUDENT**

| Sid | SName | Country |
|---|---|---|
| 1 | Aisha | Malaysia |
| 2 | Ken | Japan |
| 3 | Sara | Singapore |

a) List all course names and their categories.

b) List courses taught by teachers whose expertise is 'Data Science'.

c) Find teachers who do not teach any course.

d) List names of students who enrolled in courses after 2021.

e) List student names, course names, and teacher names.

a) List all course names and their categories.

$\pi_{cName, category}(COURSE)$

| CName | Category |
|---|---|
| Python Basics | Programming |
| Data Mining | Data Science |
| Web Design | IT |

b) List courses taught by teachers whose expertise is 'Data Science'.

$\pi_{cName}(\sigma_{Expertise = Data\ Science}(TEACHER) \bowtie COURSE)$

**TEACHER**

| Tid | Expertise |
|---|---|
| ~~1~~ | ~~Programming~~ (fail) |
| 2 | Data Science (pass) |
| ~~3~~ | ~~Cybersecurity~~ (fail) |

$\bowtie$

**COURSE**

| Tid | CName |
|---|---|
| 1 | Python Basics |
| 2 | Data Mining |
| 1 | Web Design |

=

| Tid | Expertise | CName |
|---|---|---|
| 2 | Data Science | Data Mining |

c) Find teachers who do not teach any course.

$\pi_{TName}(\pi_{Tid}(TEACHER) - \pi_{Tid}(COURSE) \bowtie TEACHER)$

**TEACHER**

| Tid |
|---|
| 1 |
| 2 |
| 3 |

−

**COURSE**

| Tid |
|---|
| 1 |
| 2 |

=

| Tid |
|---|
| 3 |

**Teacher**

$\bowtie$

| Tid | TName | Expertise |
|---|---|---|
| 3 | Prof John | Cybersecurity |

d) List names of students who enrolled in courses after 2021.

$\pi_{sName}(\sigma_{year>2021}(COURSE) \bowtie ENROll \bowtie STUDENT)$

ERROR question, "ENROll" table is not provided

e) List student names, course names, and teacher names.

$\pi_{(sName, cName, tNames}(STUDENT \bowtie ENROll \bowtie COURSE \bowtie TEACHER)$

## QUESTION 5

**ARTIST**

| Arid | ArName | Country |
|---|---|---|
| 1 | Adele | UK |
| 2 | Jay Chou | Taiwan |
| 3 | Lana Del Rey | USA |

**ALBUM**

| Alid | AlName | Year | Arid |
|---|---|---|---|
| 1 | 30 | 2021 | 1 |
| 2 | Chemtrails | 2021 | 3 |
| 3 | Fantasy | 2022 | 2 |

**TRACK**

| Tid | TName | Duration | Alid |
|---|---|---|---|
| 1 | Easy On Me | 3:44 | 1 |
| 2 | White Dress | 5:33 | 2 |
| 3 | Mojito | 3:05 | 3 |

**USER**

| Uid | UName | City |
|---|---|---|
| 1 | Hana | Kuala Lumpur |
| 2 | Rizal | Johor Bahru |
| 3 | Ken | Tokyo |

**LISTEN**

| Uid | Tid | DatePlay |
|---|---|---|
| 1 | 1 | 2024-06-01 |
| 2 | 3 | 2024-06-05 |
| 3 | 2 | 2024-06-10 |

a) List track names and their durations.

b) List track names from albums released after 2020.

c) Find artists who do not have any albums.

d) List user names who listened to tracks by artist 'Adele'.

e) List artist names, album names, and track names.

a) List track names and their durations.

$\pi_{TName, Duration}(TRACK)$

| TName | Duration |
|---|---|
| Easy On Me | 3:44 |
| White Dress | 5:33 |
| Mojito | 3:05 |

b) List track names from albums released after 2020.

$\pi_{TName}(\sigma_{Year>2020}(ALBUM) \bowtie TRACK)$

**ALBUM**

| Alid | Year |
|---|---|
| 1 | 2021 (Pass) |
| 2 | 2021 (Pass) |
| 3 | 2022 (Pass) |

$\bowtie$

**TRACK**

| Alid | TName |
|---|---|
| 1 | Easy On Me |
| 2 | White Dress |
| 3 | Mojito |

c) Find artists who do not have any albums.

$\pi_{ArName}(\pi_{Arid}(ARTIST) - \pi_{ARID}(ALBUM) \bowtie ARTIST)$

**ARTIST**

| Arid |
|---|
| 1 |
| 2 |
| 3 |

$-$

**ALBUM**

| Arid |
|---|
| 1 |
| 3 |
| 2 |

$=$

| Arid |
|---|
| NULL |

$\therefore$ No artist who do not have any albums.

d) List user names who listened to tracks by artist 'Adele'.

$\pi$ UName ($\sigma$ ArName = Adele (ARTIST) $\bowtie$ ALBUM $\bowtie$ TRACK $\bowtie$ LISTEN $\bowtie$ USER)

**ARTIST**

| Arid | ArName |
|---|---|
| 1 | Adele (Pass) |
| ~~2~~ | ~~Jay Chou~~ (Fail) |
| ~~3~~ | ~~Lana Del Rey~~ (Fail) |

$\bowtie$

**ALBUM**

| Arid | Alid |
|---|---|
| 1 | 1 |
| 3 | 2 |
| 2 | 3 |

$\bowtie$

**TRACK**

| Alid | Tid |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |

**Listen**

| Tid | Uid |
|---|---|
| 1 | 1 |
| 3 | 2 |
| 2 | 3 |

$\bowtie$

**USER**

| Uid | UName |
|---|---|
| 1 | Hana |
| 2 | Rizal |
| 3 | Ken |

$=$

| Arid | ArName | Alid | Tid | Uid | UName |
|---|---|---|---|---|---|
| 1 | Adele | 1 | 1 | 1 | Hana |

e) List artist names, album names, and track names.

$\pi$ ArName, AlName, TName (ARTIST $\bowtie$ ALBUM $\bowtie$ TRACK)

**ARTIST**

| ArName | Arid |
|---|---|
| Adele | 1 |
| Jay Chou | 2 |
| Lana Del Rey | 3 |

$\bowtie$

| Arid | AlName | Alid |
|---|---|---|
| 1 | 30 | 1 |
| 3 | Chemtrails | 2 |
| 2 | Fantasy | 3 |

$\bowtie$

| Alid | TName |
|---|---|
| 1 | Easy on Me |
| 2 | White Dress |
| 3 | Mojito |

$=$

| ArName | Arid | AlName | Alid | TName |
|---|---|---|---|---|
| Adele | 1 | 30 | 1 | Easy on Me |
| Lana Del Rey | 3 | Chemtrails | 2 | White Dress |
| Jay Chou | 2 | Fantasy | 3 | Mojito |

## QUESTION 6

Relational Schema:

EMPLOYEE (Eid, EName, City, Did)
DEPARTMENT (Did, DName, Location)
PROJECT (Pid, PName, Did)
WORKS_ON (Eid, Pid, Hours)

a) List employee names and their department IDs.

b) List projects located in the same department as employee 'Amin'.

c) Find departments that do not have any projects.

d) List employees who work more than 10 hours on any project.

e) List employee names and project names they work on.

## QUESTION 7

Relational Schema:

PATIENT (Pid, PName, Age, WardNo)
DOCTOR (Did, DName, Specialty, WardNo)
WARD (WardNo, WardName, Location)

a) List names of all doctors and their specialties.

b) List names of doctors working in the 'Surgery' ward.

c) Find ward numbers that have no patients.

d) List names of patients treated by doctors in the 'Cardiology' ward.

e) List names of doctors and patients sharing the same ward.

a) List employee names and their department IDs.

$\pi_{EName, Did}(Employee \bowtie DEPARTMENT)$

b) List projects located in the same department as employee 'Amin'.

$\pi_{PName}(\sigma_{EName=Amin}(EMPLOYEE) \bowtie Project)$

c) Find departments that do not have any projects.

$\pi_{DName}(\pi_{Did}(DEPARTMENT) - \pi_{Did}(PROJECT) \bowtie DEPARTMENT)$

d) List employees who work more than 10 hours on any project.

$\pi_{EName}(\sigma_{Hours>10}(WORKS_ON) \bowtie EMPLOYEE)$

e) List employee names and project names they work on.

$\pi_{EName, PName}(Employee \bowtie PROJECT)$

a) List names of all doctors and their specialties.

$\pi_{DName, speciality}(DOCTOR)$

b) List names of doctors working in the 'Surgery' ward.

$\pi_{DName}(\sigma_{wardName=surgery}(WARD) \bowtie DOCTOR)$

c) Find ward numbers that have no patients.

$\pi_{wardNo} - (\pi_{wardno}(WARD) - \pi_{wardno}(PATIENT) \bowtie WARD)$

d) List names of patients treated by doctors in the 'Cardiology' ward.

$\pi_{PName}(\sigma_{speciality=cardiology}(DOCTOR) \bowtie PATIENT)$

e) List names of doctors and patients sharing the same ward.

$\pi_{PName, DName}(DOCTOR \bowtie_{wardNo=wardNo} PATIENT)$

## QUESTION 8

Relational Schema:

MOVIE (Mid, Title, Year, Genre, Sid)
ACTOR (Aid, AName, Gender, Mid)
STUDIO (Sid, SName, City)

a) List movie titles and release years.

b) List titles of movies produced by 'Studio 7'.

c) Find studios that have not produced any movie.

d) List male actors who acted in movies after 2015.

e) List actor names, movie titles, and genres.

**QUESTION 9**

Relational Schema:

EMPLOYEE (Eid, EName, City, Did)
DEPARTMENT (Did, DName, Location)
PROJECT (Pid, PName, Did)
WORKS_ON (Eid, Pid, Hours)

a) List employee names and their department IDs.

b) List projects located in the same department as employee 'Amin'.

c) Find departments that do not have any projects.

d) List employees who work more than 10 hours on any project.

e) List employee names and project names they work on.

a) List movie titles and release years.

$\pi_{Title, Year}(MOVIE)$

b) List titles of movies produced by 'Studio 7'.

$\pi_{title}(\sigma_{sName = Studio\ 7}(STUDIO) \bowtie MOVIE$

c) Find studios that have not produced any movie.

$\pi_{sName}(\pi_{sid}(STUDIO) - \pi_{sid}(MOVIE) \bowtie MOVIE$

d) List male actors who acted in movies after 2015.

$\pi_{AName}(\sigma_{Year > 2015}(MOVIE) \bowtie ACTOR)$

e) List actor names, movie titles, and genres.

$\pi_{AName, Title, Genre}(MOVIE \bowtie ACTOR)$